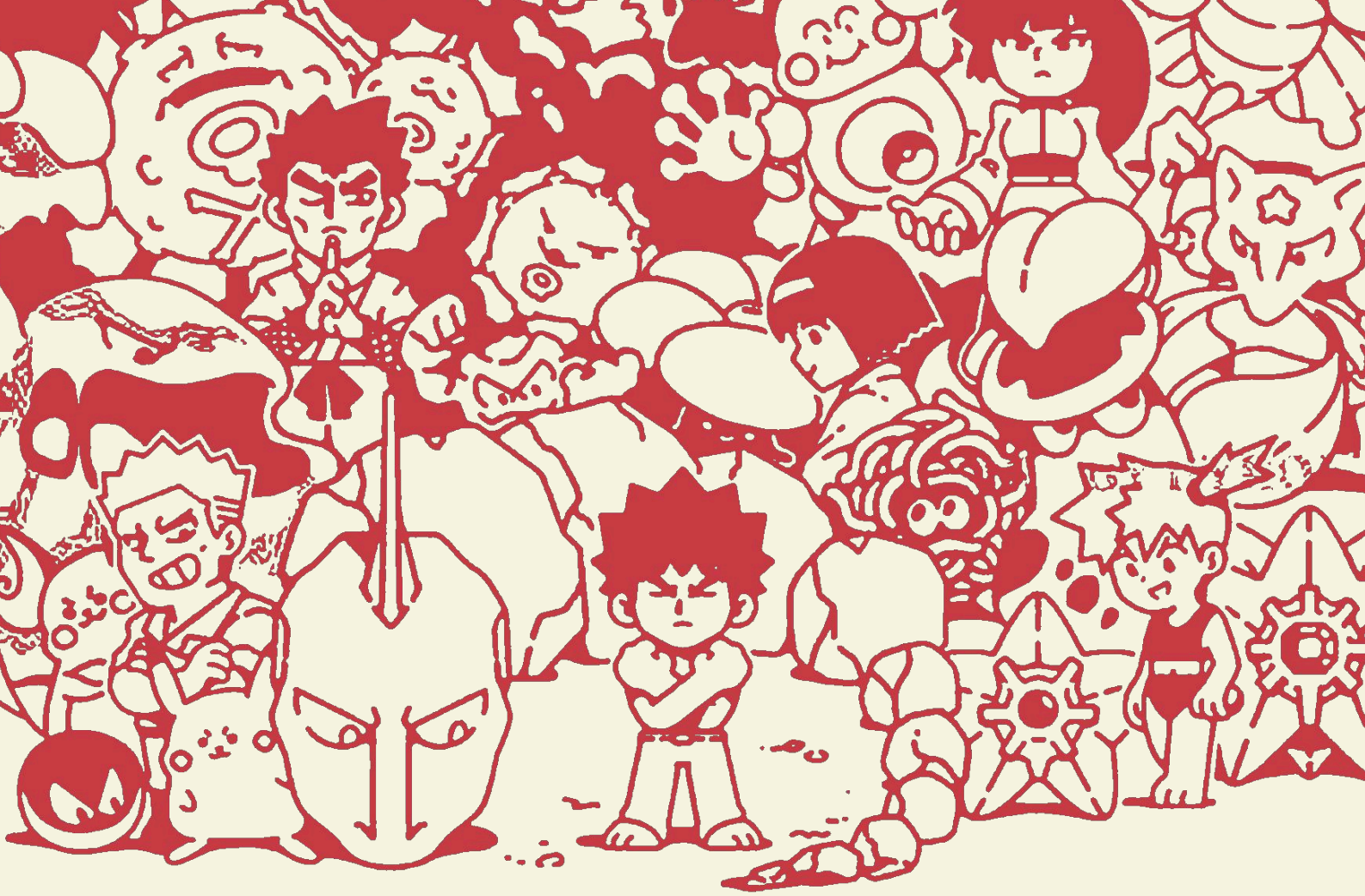
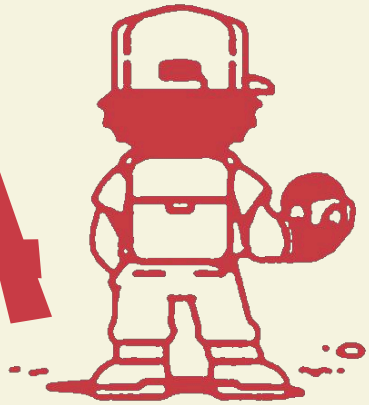




# GOTTA CODE EM ALL

## ODT CAPSTONE PROJECT

Sachit Patnaik
Tanay Dhavate

BITS Design
2025-29

# MATERIAL CHOICE

## Materials

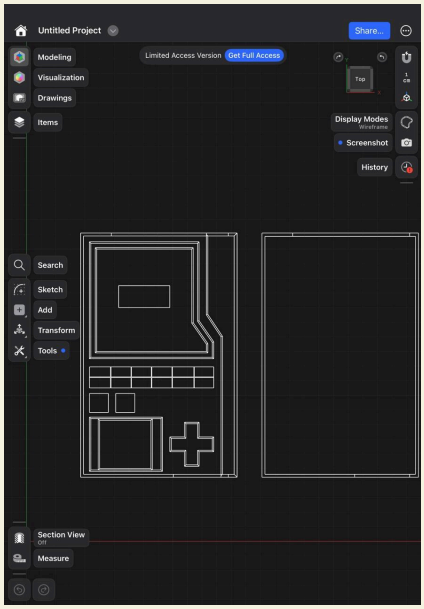
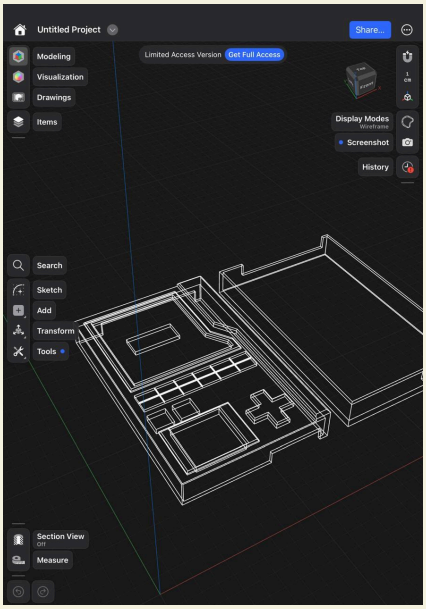
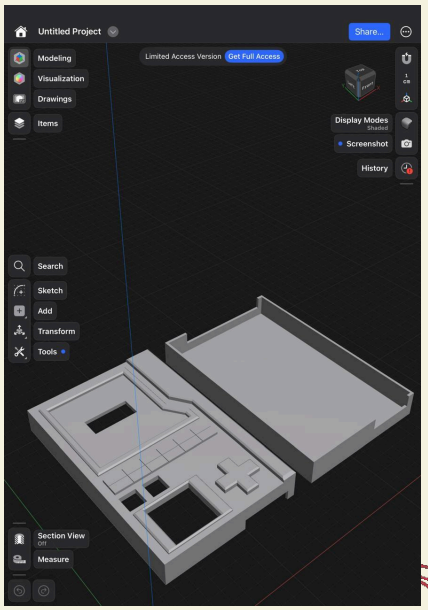
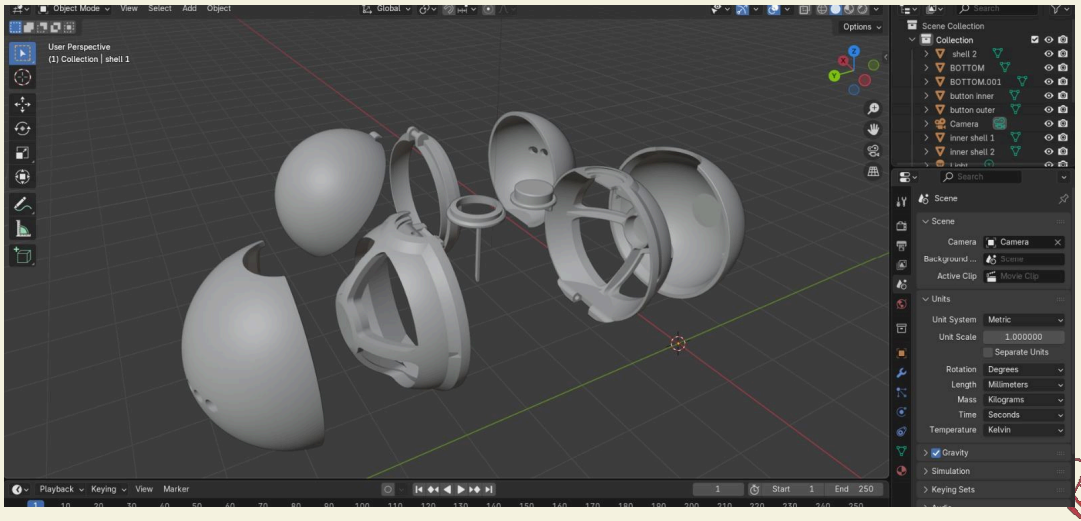
**Main Material 1- PLA**
We chose to 3D print both the Pokedex case and Pokeball using PLA.
· Structure- We chose PLA because it would give us print with accurate dimensions which was essential due to us needing to put the breadboard and most of the electronics internally. This was preferred over MDF wood due to it taking longer to assemble and it would not be as structurally sound). The overall cases turned out to be very durable and survived many drops from height and other damage.

· Adaptability- 3D printing also allowed us to change our design as the project progressed such as changing port slot sizes, height and thickness which would have been a laborious task with other materials (MDF, acrylic, metal, cardboard. We were also able to modify the physical structure easily through sanding, filing, sawing and drilling.

· Aesthetics- The original Pokedex in the shows and toys has a matte, plastic finish which we wanted to accurately represent. We did this by further sanding the prints and spray painting

**Material 2- Paper**
· Used to make the models of pokemon as cubes, done with thick glossy sheets
· Simple models, allowed us to use online cutout sheets of pokemon
· Inside was hollow, giving space to insert the unique resistor set for each pokemon as part of the ADC identification system

**Material 3- Sun board**
· Mainly used to reinforce the pokemon models to make inserting wires easier
· Chosen due it being easy to cut while remaining durable and providing adequate structural support

# FABRICATION PROCESS

2. Fabrication Steps :
A documented "Build-log" (e.g., "1. Laser cutting frame," "2. Heat-staking threaded inserts").
1. Create rough sketches of layout taking inspiration from an actual pokedex and modifying to fit the components of our project (OLED screen, push buttons, speaker, ADC reader )
2. Take dimensions of breadboard and pokemon models(made from paper cutouts) for the pokedex and DC motor dimensions for the pokeball
3. Make initial 3D models using these parameters, modified according to changes in circuitry
4. Print first models using a 3D printer and PLA material.
5. Create paper models of pokemon
6. Sanding and filing imperfections of 3d prints
7. Spray painted both models
8. Install Resistor system in each pokemon
9. Fit breadboard and wiring and some electronic components(Speaker amp, SD card reader) inside the model.
10. External electronic components(Push button, OLED, Speaker, Neopixel strip) attached to exterior of pokedex for interaction while being connected to breadboard inside
11. Connected DC motor to Pokeball and attached it to the final circuit in the pokedex
12. Secured components ran tests to ensure the whole system works